Project Sentinel: Protecting Web Security Against Future Quantum Computer Attacks

Samreen Sikander and Hasan

Aga Khan School | Researchathron

ABSTRACT

Today, most websites protect passwords and user data using security methods like RSA. However, future super-smart computers, called **quantum computers**, will easily break these old security methods using a special method called Shor's Algorithm. To fix this problem, we created **Project Sentinel**. It uses new quantum-safe security standards (NIST ML-KEM and ML-DSA) combined with a special system we designed called the *Key-Derived Masking Engine (KDMLE)*. Our tests show that Project Sentinel provides strong quantum protection (192-bit security) while keeping loading times very fast (only 22 milliseconds) and data size small (1.4 KB).

INTRODUCTION

Every time you log into a website or send private data online, encryption keeps your information safe. Traditional systems use mathematical puzzles that normal computers cannot solve. But powerful quantum computers are being built today, and they will solve these puzzles in just a few seconds.

Using *Shor's Algorithm*, a quantum computer can easily break current encryption like RSA-2048 and Elliptic Curve Cryptography (ECC). Even worse, hackers today are saving encrypted data so they can unlock it later when quantum computers become available. This threat is called **"Harvest Now, Decrypt Later."** Because of this, we must upgrade our web security to quantum-safe methods immediately.

Lattice Cryptography Diagram

Figure 1: Visualizing how lattice grid patterns create unbreakable mathematical puzzles for quantum computers.

RESEARCH PURPOSE

Scientists have already made new quantum-safe security methods (Post-Quantum Cryptography or PQC). However, these new methods send large amounts of extra data over the internet, which slows down websites and makes log-ins take too long (~38 milliseconds).

The main goal of **Project Sentinel** is to build a modern security system that stops quantum attacks completely, while keeping websites fast, light, and easy to use.

METHODS

To solve this challenge, Project Sentinel combines two main ideas:

- **Lattice-Based Encryption:** We used new official security standards (NIST FIPS 203 ML-KEM and FIPS 204 ML-DSA). Instead of basic math, these use complex multidimensional grid puzzles that quantum computers cannot solve.
- **KDMLE Speed Engine:** Verifying large lattice keys on every single click slows down the website. Our KDMLE engine verifies the user once and then creates small, temporary security tokens for fast browsing.

- **Testing Setup:** We tested 1,000 log-in requests across three systems: Old RSA, Standard PQC, and our optimized Project Sentinel.

Performance Latency Benchmarks

Figure 2: Chart comparing speed, data size, and quantum safety across different systems.

RESULTS

Our testing across 1,000 log-in sessions showed the following results:

Security System	Log-in Speed (Latency)	Data Size (Payload)	Quantum Protection
Old RSA-2048 + JWT	8 ms (Fast)	0.4 KB (Small)	No Protection (Unsafe)
Standard PQC Protocol	38 ms (Slow)	4.6 KB (Large)	192-bit Strong Quantum Safety
Project Sentinel (KDMLE)	**22 ms (Balanced)**	**1.4 KB (Optimized)**	**192-bit Strong Quantum Safety**

Project Sentinel reduced the data size from 4.6 KB down to 1.4 KB and improved the log-in speed from 38 ms to 22 ms, while keeping the exact same high level of security.

DISCUSSION

Our test results show that Project Sentinel successfully solves the big problem of post-quantum security. Old RSA systems are fast (8 ms) but offer no protection against quantum computers. Standard PQC is secure but slows down websites too much (38 ms).

Project Sentinel finds the perfect middle ground: adding just 14 milliseconds over old systems (which users won't even notice) while providing full protection against quantum attacks. It also allows future updates easily if security standards change.

REFERENCES

- NIST. (2024). *FIPS 203: Key-Encapsulation Mechanism Standard (ML-KEM)*. U.S. Department of Commerce.
- NIST. (2024). FIPS 204: *Digital Signature Algorithm Standard (ML-DSA)*. U.S. Department of Commerce.
- Shor, P. W. (1994). *Algorithms for quantum computation: discrete logarithms and factoring*. IEEE FOCS.
- Bernstein, D. J., & Lange, T. (2017). *Post-quantum cryptography overview*. Nature, 549, 188-194.